

Table 4. Interconnect of peripherals (continued)

Interconnect source	Interconnect destination	Interconnect action	Run	Sleep	Stop
GPIO	TIM1,2,3	External trigger	Y	Y	-
	ADC	Conversion external trigger	Y	Y	-

3.9 Clocks and startup

The clock controller distributes the clocks coming from different oscillators to the core and the peripherals. It also manages clock gating for low-power modes and ensures clock robustness. It features:

- **Clock prescaler:** to get the best trade-off between speed and current consumption, the clock frequency to the CPU and peripherals can be adjusted by a programmable prescaler
- **Safe clock switching:** clock sources can be changed safely on the fly in run mode through a configuration register.
- **Clock management:** to reduce power consumption, the clock controller can stop the clock to the core, individual peripherals or memory.
- **System clock source:** the following clock sources can deliver SYSCLK system clock:
 - 4-48 MHz high-speed oscillator with external crystal or ceramic resonator (HSE). The HSE can also be configured in bypass mode for an external clock.
 - 48 MHz high-speed internal RC oscillators (HSI48 and HSIUSB48), trimmable by software.
 - 32.768 kHz low-speed oscillator with external crystal (LSE), supporting two drive capability modes. The LSE can also be configured in bypass mode for using an external clock.
 - 32 kHz low-speed internal RC oscillator (LSI) with $\pm 5\%$ accuracy, also used to clock an independent watchdog.
- **Peripheral clock sources:** several peripherals (I2S, USART1, I2C1, ADC, USB FS) can operate with a clock source independent of the system clock. The HSIUSB48 RC oscillator associated with CRS allows the USB FS peripheral to operate without requiring an external resonator, as a USB device.
- **Clock security system (CSS):** in the event of HSE or LSE clock failure, the system clock is automatically switched to HSI48 or LSI, respectively. If enabled, a software interrupt is generated. The CCS feature can be enabled by software.
- **Clock output:**
 - **MCO and MCO2 (microcontroller clock output)** provides one of the internal clocks for external use by the application.
 - **LSCO (low speed clock output)** provides LSI or LSE in all low-power modes.

Several prescalers allow the application to configure AHB and APB domain clock frequencies, 48 MHz at maximum.